A PROJECT TITLE

ON

Insider Threat Detection Using Elastic Net Regularized Logistic Regression

MACHINE LEARNING

BACHELOR OF TECHNOLOGY

in

A.Y.: 2026-27

by

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Deparment Of Computer Science Engineering

Name of Title: **Insider Threat Detection Using Elastic Net Regularized Logistic Regression**

1. Abstract:

Insider Threat Detection Using Elastic Net Regularized Logistic Regression

Insider threats are one of the most challenging cybersecurity issues faced by organizations today. Unlike external cyberattacks, insider threats originate from individuals who have authorized access to organizational resources, such as employees, contractors, or business partners. These threats can lead to data breaches, financial losses, and damage to an organization's reputation. Detecting malicious insider activities is difficult because insider behavior often resembles normal user activities.

This project proposes an Insider Threat Detection System using Elastic Net Regularized Logistic Regression, a machine learning approach that combines the strengths of L1 (Lasso) and L2 (Ridge) regularization techniques. The model analyzes employee behavioral data, including login patterns, file access history, USB device usage, email activities, and failed login attempts, to classify users as either normal or potential insider threats.

The collected dataset is preprocessed by handling missing values, encoding categorical features, and normalizing numerical data to improve model performance. Elastic Net regularization helps select the most relevant features while reducing overfitting, resulting in a more accurate and robust prediction model. The trained model is evaluated using performance metrics such as accuracy, precision, recall, F1-score, and confusion matrix.

The proposed system enables organizations to identify suspicious user behavior at an early stage, allowing security teams to take preventive actions before significant damage occurs. By integrating machine learning with cybersecurity, this project provides an efficient, scalable, and reliable solution for detecting insider threats and enhancing organizational security.

2. INTRODUCTION:

With the rapid growth of digital technologies, organizations increasingly rely on computer networks and information systems to manage sensitive data and daily operations. While traditional cybersecurity measures focus on protecting systems from external attacks, threats originating from within the organization have become a major concern. These threats, known as insider threats, are caused by employees, contractors, or other authorized users who intentionally or unintentionally misuse their access privileges, leading to data breaches financial losses, and reputational damage.

Detecting insider threats is challenging because insiders already possess legitimate access to organizational resources, making malicious activities difficult to distinguish from normal user behavior. Conventional rule-based security systems often fail to identify such threats due to the complexity and dynamic nature of employee activities.

Machine Learning (ML) provides an intelligent approach to detecting insider threats by analyzing user behavior patterns and identifying suspicious activities. This project proposes an Insider Threat Detection System using Elastic Net Regularized Logistic Regression, a supervised machine learning algorithm that combines the advantages of L1 (Lasso) and L2 (Ridge) regularization. Elastic Net improves prediction accuracy by selecting the most relevant features while reducing overfitting.

The system analyzes employee activities such as login time, file access, email usage, USB device usage, and failed login attempts to classify users as either Normal or Potential Inside Threats. By providing early detection of suspicious behavior, the proposed system enables organizations to take preventive actions before significant security incidents occur.

This project demonstrates how machine learning techniques can enhance cybersecurity by improving the efficiency, accuracy, and scalability of insider threat detection systems.

3. SOFTWARE REQUIREMENTS:

Programming Language

- Python 3.10 or above

Development Environment

- Visual Studio Code (VS Code)
- Jupyter Notebook (Optional)

Machine Learning Libraries

- Pandas

- NumPy
- Scikit-learn
- Matplotlib
- Seaborn
- Joblib

Framework

- Streamlit (for the user interface, if used)

Dataset

- CERT Insider Threat Dataset / Kaggle Insider Threat Dataset (CSV format)

Database

- CSV File

Operating System

- Windows 10/11
- Ubuntu Linux (Optional)

Machine Learning Algorithm

- Logistic Regression with Elastic Net Regularization

Version Control (Optional)

- Git

Signature of the Guide

Course Instructor